Executive Summary: UI, Dashboard & Search

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Executive Dashboard Metrics, Global Command Search, Calculator Navigation, Project Lifecycle & Direct Route Access (IDs 25, 26, 33, 34, 37, and 40).

ID	Checklist Item	Severity	Audit Result	Action Required
ID 25	Dashboard Metrics (Real-Time Aggregations)	High	PASS (100%)	None - Ventilation airflow & compliance stats active
ID 26	Global Search (Command Palette Cmd+K / Ctrl+K)	High	PASS (100%)	None - Calculator & space finder operational
ID 33	Calculator View Navigation	High	PASS (100%)	None - Direct module routing verified across 25+ calcs
ID 34	Projects - Safe Deletion Handling	High	PASS (100%)	None - Cascade cleanup in DB and local caches active
ID 37	Direct URL / Feature Route Access Gating	High	PASS (100%)	None - AuthGuard router checks enforce active session
ID 40	Organizations Page Search & Filtering	Medium	PASS (100%)	None - Real-time client & server filtering operational

1. Detailed Architecture Verification & Findings

Item ID	Architecture Component	Technical Verification & Evidence
ID 25	Executive Dashboard File: frontend/src/routes/dashboard.tsx Component: DashboardPage	EVIDENCE: Aggregates total fresh intake (L/s), extract air, compliance score (%), critical pressure imbalance warnings, and recent calculations with automatic refresh on project switch.
ID 26	Global Command Palette (Cmd+K / Ctrl+K) File: frontend/src/components/workspace/AppShell.tsx	EVIDENCE: Quick-search dialog indexes all MEP modules, spaces, and project entities. Supports keyboard navigation (Arrow Up/Down, Enter) for instantaneous routing.
IDs 33 & 37	Calculator View & Route Protection File: frontend/src/routes/__root.tsx Component: AuthGuard	EVIDENCE: Sidebar links route cleanly to individual unlocked calculators. Direct deep-link URL entry is authenticated and gated; unauthenticated visitors are redirected to `/login`.
ID 34	Project Deletion & Cleanup File: frontend/src/routes/projects.tsx Endpoint: POST /api/projects/delete	EVIDENCE: Confirms deletion modal, removes project records, cascade deletes associations in `project_users` and `project_teams`, and clears cached local stores.
ID 40	Organization Search Filtering File: frontend/src/routes/admin.tsx	EVIDENCE: Real-time search filters organization lists by name, plan tier, and active license key with instantaneous UI updates.

2. UI Interaction & Navigation Architecture

The following diagram outlines the global search indexing, dashboard analytics aggregation, and route protection architecture:

1. Executive Dashboard Aggregator

- Computes active ventilation balance: sum(freshAir), sum(extractAir), imbalance thresholds.
- Computes ASHRAE 62.1 compliance score: 100 - (non-compliant spaces / total spaces * 100).

2. Global Command Palette (Cmd+K / Ctrl+K)

- In-memory search index covers all 25+ calculators, space codes, and project categories.
- Direct shortcut listener (`Cmd+K` / `Ctrl+K`) activates modal from any application screen.

3. Route Guard & Project Deletion Safety

- TanStack Router + AuthGuard enforce authorization before component mounting.
- Project deletion clears references from database and switches active workspace to fallback safely.

3. Audit Conclusion & Compliance Status

All 6 checklist requirements (IDs 25, 26, 33, 34, 37, and 40) under UI, Dashboard, Navigation & Search are fully implemented, verified, and 100% compliant with modern responsive web standards. No code modifications are required for this module.